

Sofiia Tretiak

AI and Python Engineer | Medical Informatics

tretiak.sofiia03@gmail.com · Vienna, Austria · [linkedin.com/in/sofia-tretiak](https://www.linkedin.com/in/sofia-tretiak) · github.com/TSofi

Profile

I build AI and data products end to end and put them in front of real users. My own platform answers medicines questions from six national drug registers with cited sources; before that I trained deep learning models on EEG data in a neuroengineering lab, and I spent a year at Volvo standing between users and engineers - finding out what people actually needed, checking data with SQL, testing API integrations and writing the documentation the team kept reusing. I also evaluate LLM output professionally, so I treat model answers as something to verify. Medical informatics is my background and health tech is where I want to stay, but the engineering travels anywhere. English C1, Ukrainian native, Polish C1, German A2.

Projects

PharmaRAG - multilingual medicines assistant | personal project, live and open source

- Built and shipped the whole system alone: ingestion from six national drug registers, cleaning and normalisation, multilingual embeddings and ~400k chunks in a Qdrant vector database, retrieval with hybrid re-ranking
- LLM layer with Groq and a Gemini fallback: answers with source citations, safety tiers for medical content, six answer languages
- FastAPI on Render, React on Vercel, 36 automated tests, answers in 2-4 seconds; live and open source

Technologies: Python, FastAPI, Qdrant, embeddings, Groq and Gemini APIs, React, Render, Vercel, pytest

GEN-DOCK - molecular docking pipeline with clinical data mapping | MSc project, FH Technikum Wien

- Containerised docking pipeline: receptor and ligand preparation, Vina runs, scripted parsing of binding affinities into structured data, results mapped to HL7 FHIR resources for a clinical decision support view

Technologies: Python, AutoDock Vina, RCSB PDB, PubChem, PyMOL, Docker, HL7 FHIR R4

EyeSpy - Real-Time Emotion and Fatigue Detection | Solvro Summer Challenge 2025

- ResNet18 on FER-2013 with per-class failure analysis, integrated into a real-time video pipeline with MediaPipe landmark tracking (EAR/MAR fatigue metrics)

Technologies: Python, PyTorch, ResNet18, OpenCV, MediaPipe

Career Shark - Full-Stack Web Application | BEST Hackathon, Top 10

- Team lead of four; 7th out of about 50 teams with a full-stack product (React, FastAPI, MongoDB) built and deployed in 24 hours

Technologies: React, FastAPI, Python, MongoDB

Experience

Research Trainee, Neuroengineering Department

METAM, Karadeniz Technical University | Trabzon, Türkiye | Jun 2026 – Aug 2026

- Designed, trained and evaluated GRU-Transformer models in PyTorch for 9-class emotion recognition from multichannel EEG
- Engineered differential-entropy and functional-connectivity features; ran ablation studies on data volume, channels and signal duration

AI Output Reviewer and Data Annotator

Outlier and Alignerr | Remote | Mar 2026 – Present

- Evaluate LLM outputs across text, code and structured data against detailed rubrics: hallucinations, edge cases and systematic failure modes

Software Engineer Intern, Digital Product Team

Volvo | Wrocław, Poland | May 2025 – May 2026

- Clarified what users actually needed and turned it into clear tickets for engineering; handled 5-15 requests a day from first to third level in ServiceNow and Azure DevOps
- Ran SQL queries and prepared views against production databases to validate data and confirm root causes; tested REST API integrations
- Worked on a company-wide migration of user and access management across several countries; documented it and presented progress to stakeholders

Skills

AI and LLM: RAG architectures, embeddings and vector databases (Qdrant), LLM API integration (Groq, Gemini, Claude), prompt design, retrieval and output evaluation, AI-assisted development

Machine Learning: PyTorch, Transformer and GRU models, CNNs (ResNet), feature engineering, ablation studies, evaluation protocols, scikit-learn, OpenCV, MediaPipe

Engineering: Python, FastAPI, SQL, React, Docker, Git, REST APIs, JSON, pytest, cloud deployment (Render, Vercel), Azure DevOps, Jira

Analysis and Delivery: requirements clarification with non-technical stakeholders, data validation and reporting, process and system documentation, test cases, working between business and engineering teams

Domain: medical informatics, HL7 FHIR, clinical and regulatory data, bioinformatics (molecular docking, PDB structures), biomedical signal processing

Education & Certifications

Master of Science, Medical Engineering and eHealth

UAS Technikum Wien, Vienna | 2026 – 2028

Bachelor of Engineering, Medical Informatics and Bioengineering

Wrocław University of Science and Technology | 2022 – 2026

- Oracle Database 19c Administration Professional – in progress

Languages

Ukrainian (native), English (C1), Polish (C1), German (A2, actively learning)